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GRU-Based Deep Learning Framework for Real-Time, Accurate, and Scalable UAV Trajectory Prediction

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Abstract: Trajectory prediction is critical for ensuring the safety, reliability, and scalability of Unmanned Aerial Vehicle (UAV) in urban environments. Despite advances in deep learning, existing methods often struggle with dynamic UAV conditions, such as rapid directional changes and limited forecasting horizons, while lacking comprehensive real-time validation and generalization capabilities. This study addresses these challenges by proposing a gated recurrent unit (GRU)-based deep learning framework optimized through Look_Back and Forward_Length labeling to capture complex temporal patterns. The model demonstrated state-of-the-art performance, surpassing existing unmanned aerial vehicles (UAV) and aircraft trajectory prediction approaches, including FlightBERT++, in terms of both accuracy and robustness. It achieved reliable long-range predictions up to 4 s, and its real-time feasibility was validated due to its efficient resource utilization. The model's generalization capability was confirmed through evaluations on two independent UAV datasets, where it consistently predicted unseen trajectories with high accuracy. These findings highlight the model's ability to handle rapid maneuvers, extend prediction horizons, and generalize across platforms. This work establishes a robust trajectory prediction framework with practical applications in collision avoidance, mission planning, and anti-drone systems, paving the way for safer and more scalable UAV operations.

Keywords: unmanned aerial vehicle; UAV trajectory prediction; deep learning; real-time



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1. Introduction

1.1. UAV Trajectory Prediction: Background and Importance

Unmanned aerial vehicles (UAVs) or drones have become essential in logistics, agriculture, disaster management, environmental monitoring, and many other fields [1,2]. Recent advances in sensor miniaturization and communication technologies have enabled these autonomous systems to operate properly in both urban and remote areas. For example, UAVs increase the speed of last-mile delivery by facilitating goods transport, while real-time aerial data can be used in precision agriculture to improve irrigation and crop health monitoring [3]. In disaster response, drones can improve situational awareness through the collection and transmission of real-time data, enhancing the speed of rescue operations [4].

However, UAVs face challenges in complex and unpredictable conditions requiring accurate trajectory prediction. Relevant constraints include changes in weather, regulatory changes, and other unforeseen obstacles and airspaces; therefore, adaptive solutions are required. In particular, in urban search and rescue air missions, the limited mobility of

UAVs flying in congested areas makes it difficult to adjust their path in response to changes in the terrain or the presence of obstacles [5,6]. Apart from basic collision avoidance, better prediction techniques can increase the energy efficiency and, therefore, the flight range and downtime of UAVs [7]. In addition, the prediction performance has a direct impact on the success of time-critical operations (e.g., medical supply deliveries) [8].

Conventional methods for trajectory planning and rule-based methods for navigation under relatively stable conditions are well established [9]. However, modern UAV missions often take place in data-intensive, dynamic, and non-linear environments, where drones may exhibit a variety of behaviors. Furthermore, urban corridors, restricted airspaces, and intermittent external data streams (e.g., from sensors or satellite imagery) add complexity to real-time decision-making. Therefore, there is an increasing need for better and more adaptive methods for predicting the trajectories of UAVs that can incorporate various datasets, follow non-linear patterns, and respond well to changes in the environment—capabilities that existing methods may not be able to offer.

1.2. Traditional Approaches to UAV Trajectory Prediction: Research Trends and Limitations

Prior to the development of deep learning techniques, UAV trajectory prediction depended mainly on conventional trajectory planning and fundamental machine learning methods. To find feasible trajectories, algorithms such as Dijkstra, A* and Rapidly exploring Random Trees (RRT) have been used to model the environment as a static or partially known map [10]. Such approaches are quite effective under well-organized or less dynamic operating conditions [11]; however, their fundamental assumptions may not adapt well to scenarios involving dynamic or unpredictable factors, such as moving obstacles and weather changes, resulting in less accurate—or even sub-optimal—trajectories in real-world scenarios.

To enforce constraints such as obstacle avoidance or no-fly zones, some researchers have used optimization-based frameworks (e.g., Mixed Integer Linear Programming) [12]. Although these methods can generate high-precision flight trajectories for a given scenario, they become computationally expensive or infeasible when considering large, rapidly changing state spaces [13]. Similarly, early probabilistic approaches, including Bayesian planners and Markov Decision Processes, offered robustness against sensor noise but were not very effective in highly unpredictable environments [14,15].

At the same time, early machine learning-based methods had limited capacity to adapt to shifting environmental conditions, as they typically relied on simpler function approximators such as decision trees or support vector machines [16]. These methods required extensive data and thorough pre-processing and, although they introduced some learning capabilities, they tended to handle out-of-distribution samples poorly. Moreover, their prediction accuracy was constrained by rigid assumptions about the environment and a lack of representative training data. While these conventional approaches contributed to understanding how UAV trajectories can be predicted, they also revealed a need for more advanced techniques that are capable of adapting to evolving conditions and large-scale datasets. As these traditional methods applied strict models that often failed in complex tasks, deep learning approaches—which offer the ability to model non-linear relationships and leverage extensive data—emerged as a promising solution for accurate and contextual trajectory forecasting.

1.3. Deep Learning Approaches for UAV Trajectory Prediction

In order to investigate recent trends in deep learning-based Unmanned Aerial Vehicle (UAV) trajectory prediction, we surveyed a set of high-performing contemporary studies. Additionally, to gain further insight into this research direction, we examined

recent works on deep learning-based aircraft trajectory prediction, as these studies have been conducted for a longer period and may offer lessons that are applicable to UAV contexts. A brief summary of these studies is provided in Table 1, which also highlights important elements such as the dataset size, sampling intervals, deep learning architectures, performance measures, and the flying domain (UAV or airplane). We observed clear trends in data volume, horizon length, and real-time feasibility claims when comparing UAV-focused papers (e.g., [17–20]) with aircraft-oriented works (e.g., [21–24]). These differences are primarily due to the different operational contexts of low-altitude, rapidly maneuvering drones versus higher-altitude, relatively steadier aircraft routes.

Table 1. Summary of recent (within the past five years) UAV and aircraft trajectory prediction studies, comparing flight type, dataset attributes (number of routes, average data points per route, data interval), deep learning models, evaluation metrics, real-time feasibility, and maximum prediction horizons. Each work highlights distinct performance metrics (e.g., root mean squared error (RMSE), mean absolute error (MAE), Euclidean distance) at various forecasting steps ($t + 1$, $t + \text{MAX}$), illustrating the breadth of approaches and reported accuracy in short- and long-range predictions.

Study	Flight Type	Dataset			Deep Model	Evaluation					
		No. of Routes	Avg. Data Points per Route	Data Interval (s)		Evaluation Metric	Real Time?	Comparison Model	MAX Prediction Point (Step)	t + MAX Prediction (RMSE)	t + 1 Prediction (RMSE)
Shu et al. [17] (2021)	UAV	25	-	-	SBULSTM	MSE	-	-	-	-	0.374 m (Mean Distance)
Kant et al. [18] (2022)		13	720	10	LSTM-Autoencoder	MAE, MSE, RMSE	O	Regression, LSTM	-	-	0.1104
Chen et al. [19] (2023)		20	497.55	0.2	EC-Bessel-BiLSTM	RMSE	O	Bi-LSTM, LSTM, CNN, BP	1.2 s (6)	Lat: 0.4724 Lon: 0.3509 Alt: 0.4143	Lat: 0.0331 Lon: 0.0271 Alt: 0.0287
Palamas [20] (2023)		182	1250	0.1	LSTM	Euclidean, Precision, Recall, F1-score	O	-	0.3 (3)	8.0021 (Euclidean)	2.5521 (Euclidean)
Ma et al. [21] (2020)	Aircraft	1	-	1	CNN-LSTM	RMSE, MAE, MAPE	O	LSTM, BP	5 s (5)	10.192	36.024
Guo, D. et al. [22] (2023)		144,605	-	20	FlightBERT	MAE, RMSE	O	LSTM, Bi-LSTM, Kalman-Filter	300 s (15)	Lat: 0.1067 Lon: 0.1199 Alt: 34.02	Lat: 0.0486 Lon: 0.0558 Alt: 10.59
Guo, D. et al. [23] (2024)		8643	-	20	FlightBERT++	RMSE, MAE, MAPE, MDE	O	LSTM, Transformer, Kalman-Filter, FlightBERT, LSTM + Attention	300 s (15)	Lat: 0.0326 Lon: 0.0265 Alt: 22.89	Lat: 0.0115 Lon: 0.0037 Alt: 12.07
Lan Wu et al. [24] (2024)		274,605	140	10	FlightPatchNet	RMSE, MAE	O	LSTM, Bi-LSTM, CNN-LSTM, FlightBERT, FlightBERT++	150 s (15)	Lat: 0.0117 Lon: 0.0157 Alt: 244.34	Lat: 0.00064 Lon: 0.00087 Alt: 123.78

1.4. Comparative Insights: Trajectory Prediction of Conventional Aircraft and Unmanned Aerial Vehicles

Although both conventional aircraft and UAVs depend on sophisticated flight control and navigation systems, each domain poses distinct obstacles to trajectory prediction. While UAVs typically operate at low altitudes, exhibit high-frequency flight dynamics, and maneuver swiftly, in contrast, fixed-wing aircraft typically maintain steadier routes at higher elevations. Recent studies revealed distinct differences in datasets, architectures, and metrics due to these distinctions.

Typically, UAV research involves smaller-scale datasets, typically consisting of only a few hundred data points per route or tens of flight routes. Due to their low-altitude, high-maneuverability context, UAVs often gather short-interval data (e.g., sampling rates below 1 s). This setup naturally favors long short-term memory (LSTM)-based deep learning models with relatively fewer parameters, which are more suitable for handling abrupt heading changes and frequent directional shifts. Almost all studies have used the RMSE to measure prediction accuracy, and many compared their results solely against other time-series-specific networks (e.g., Bi-LSTM, convolutional neural networks (CNNs), or basic recurrent neural networks (RNNs)). While some UAV studies have claimed real-time capability by referencing sub-0.1 s inference times, they typically offer only single-step or short multi-step horizons (up to about 1.2 s), limiting their applicability for longer-range planning and obstacle avoidance.

In contrast, conventional aircraft trajectory prediction research benefits from larger datasets, often ranging into tens or hundreds of thousands of flight routes. The sampling intervals also tend to be longer (e.g., 10 or 20 s), accommodating heavier parameter networks such as transformer-based architectures (e.g., FlightBERT, FlightBERT++). While these studies frequently compare multiple deep learning methods, including CNN–LSTM hybrids and attention-based models, direct cross-model comparisons are still somewhat limited; most have focused on a handful of network variants, rather than diverse computational baselines. Nevertheless, conventional aircraft routes allow for far longer forecasting windows—sometimes extending hundreds of seconds—as the steadier, higher-altitude environment reduces rapid directional changes. As a result, aircraft research frequently produces forecasts with greater precision than those in UAV contexts, regardless of whether they are short- or long-term.

In conclusion, aircraft trajectory prediction research is characterized by the ability to accommodate longer intervals and larger data repositories, which often results in more parameter-heavy architectures that are particularly effective for long-range forecasting. In contrast, UAV trajectory prediction research is required to manage rapid maneuvering and high-frequency sampling with comparatively smaller datasets. However, aircraft research, which has developed over an extended period, may provide valuable insights (e.g., multi-step or parallel prediction strategies) to assist UAV-based approaches in overcoming the current limitations in terms of data volume, sampling intervals, and maximum prediction horizons, despite the inherent differences.

1.5. Outstanding Challenges and Our Contributions

Prior to the emergence of deep learning, early machine learning techniques frequently focused on more rudimentary function approximators. Despite significant progress in UAV trajectory prediction, some critical difficulties persist, hindering the efficacy and usability of current approaches. The following deficiencies persist in the existing literature:

- **Lower Prediction Accuracy in Frequent UAV Maneuvers:**

UAV trajectory prediction methods typically yield lower accuracy than those developed for fixed-wing or higher-altitude aircraft, largely because UAVs often rapidly change direction at low altitudes to avoid obstacles. Such frequent, abrupt maneuvers diverge sharply from the steadier flight patterns that many existing models expect, complicating their ability to capture sudden trajectory deviations. Consequently, while UAV-focused deep learning techniques continue to advance, they still lag behind the performance achieved in aircraft trajectory forecasting, indicating a need for further development and optimization in UAV-specific applications.

- **Primarily Short-Term Predictions:**

Multiple investigations have indicated robust performance solely within short prediction horizons, with accuracy swiftly diminishing beyond these periods. This limitation obstructs sophisticated operations (e.g., long-distance cargo transit, continuous surveillance) that depend on anticipatory decision-making and long-range trajectory analysis.

- **Inadequate Real-Time Validation:**

Some studies provide no evidence of real-time trajectory prediction at all, while others merely state that their inference time is shorter than the sensor sampling interval. However, a more logical demonstration is needed to establish genuine real-time feasibility, especially under the broader pipeline delays and variable flight conditions encountered in practice.

- **Insufficient Evidence of Cross-UAV Generalization:**

Many studies train and test on the same UAV datasets, leaving it unclear whether their models can adapt to distinct UAV platforms or flight logs. Consequently, it remains uncertain how well these approaches generalize when faced with different flight dynamics or sensor configurations. Without comprehensive cross-dataset or cross-platform evaluations, questions about scalability and reliability persist, particularly in scenarios where UAVs exhibit diverse operational characteristics.

This work introduces an innovative deep learning methodology that is designed to improve predictive accuracy, broaden forecasting horizons, possess rigorously validated near-real-time performance, and enable cross-platform adaptability. Our work provides the following contributions:

1. **High-Performance Deep Learning Model for UAV Trajectory Prediction**

We create a customized network architecture specifically designed for UAV trajectory prediction, surpassing current methods in terms of overall accuracy, especially in dynamic flying scenarios characterized by sudden directional changes and low-altitude maneuvers.

2. **Prolonged Forecasting Horizons**

In addition to attaining precision within brief intervals, our methodology ensures strong predictive efficacy for distant future positions, allowing UAVs to forecast their trajectories further ahead. The ability for long-range prediction is essential for sophisticated mission planning, obstacle evasion, and resource allocation under dynamic flight conditions.

3. **Comprehensive Evidence of Real-Time Viability**

We establish real-time feasibility by demonstrating that our model consistently predicts future states in a significantly shorter period than the data sampling interval. In addition, we quantify CPU and GPU usage during inference to provide a logical, data-driven basis for near-real-time applicability. This approach ensures our method meets practical timing constraints without compromising predictive accuracy.

4. Cross-UAV Assessment for Generalization

To validate its scalability, we evaluate our approach using supplementary datasets obtained from different UAVs within the same category; however, these UAVs are not similar to the training platform. The strong performance of the model on these unfamiliar, yet comparable aircraft highlights its adaptability across various missions and hardware configurations.

The subsequent sections of this work are structured as follows. Section 2 presents the trajectory dataset and details the deep learning model utilized in this study. Section 3 delineates the experimental design and performance evaluation, specifying the assessment of predicted accuracy and associated indicators. Section 4 presents an extensive discussion, addresses potential applications, highlights key contributions, and offers a direct performance comparison with recent aircraft-based models. Section 5 then summarizes this study's principal findings, acknowledges its limitations, and proposes avenues for subsequent advancements.

2. Materials and Methods

2.1. Overall Workflow

To systematically develop and validate our deep learning-based trajectory prediction approach, we designed a multi-stage workflow that spans data acquisition, labeling, model development, and performance evaluation (Figure 1). The following steps outline the major components:

A. Data Collection

We sourced an open dataset comprising 81,169 UAV flight trajectories, each containing latitude, longitude, and altitude (x, y, z) measurements. As the raw data were already of sufficient quality—featuring consistent time stamps and minimal missing values—no extensive pre-processing was required beyond general integrity checks.

B. Data Labeling

To enable multi-step forecasting, we introduced two parameters: *Look_Back* and *Forward_length*. *Look_Back* specifies how many past observations (time steps) are provided to the model as input, while *Forward_length* indicates how many future points (e.g., $t + 1, t + 2, \dots$) the network should predict. Using these parameters, we convert each flight record into input-label pairs for both training and testing. This segmentation ensures that the model can learn from historical trends while forecasting multiple time steps ahead.

C. Deep Learning Model and Real-Time Optimization

We employ a gated recurrent unit (GRU)-based architecture tailored to UAV trajectory prediction, taking into account rapid heading changes and potentially low-altitude operations. Various layer depths, hidden dimensions, and activation functions are tested to maximize the accuracy of the resulting model while keeping the computational overhead low.

- Hyperparameter tuning: By iteratively adjusting the batch size, learning rate, and number of hidden units, we sought a balance between predictive precision and inference speed.
- Near-real-time constraints: Model complexity was minimized to ensure that sub-0.1 s (or similar) inference times could be maintained, thereby supporting real-time or near-real-time applications.

D. Performance Evaluation

Finally, we assess the model using RMSE across latitude, longitude, and altitude predictions. We also visualize the predicted trajectories against ground-truth logs to examine both short-horizon and extended forecasts. In addition, we gauge the feasibility of its real-time performance by measuring the average inference latency under typical conditions. To confirm its generalization capabilities, the final model is further tested on data from distinct UAV types, allowing us to explore its cross-platform adaptability.

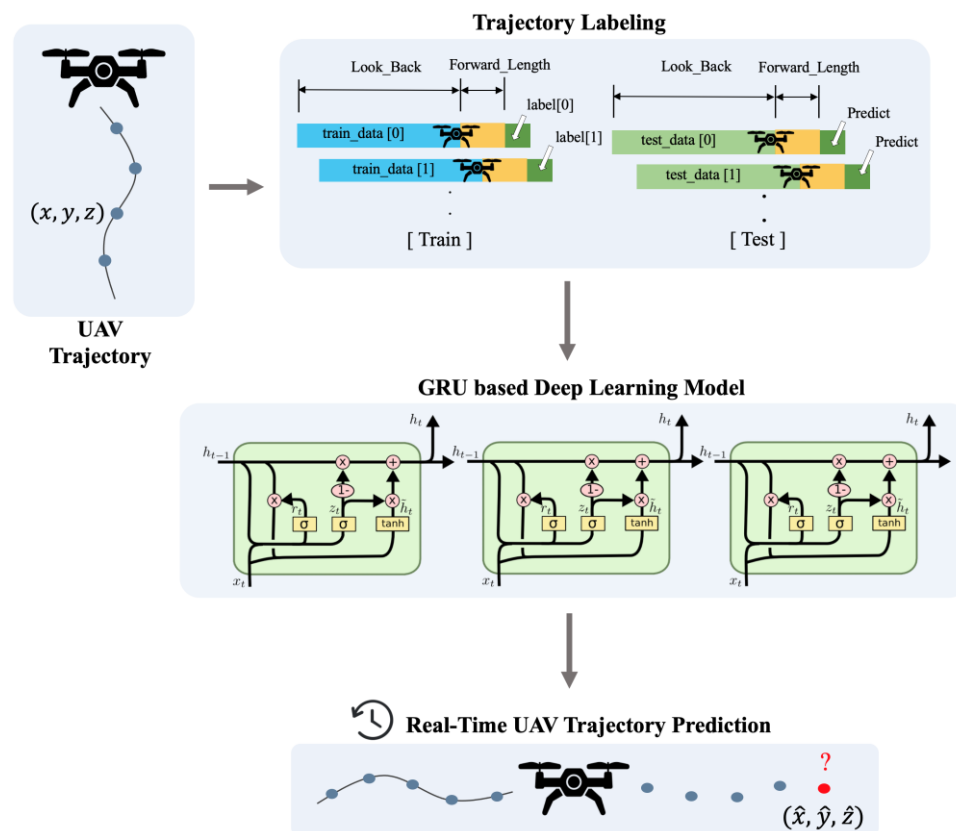


Figure 1. Overall workflow of the proposed real-time UAV trajectory prediction framework. The raw flight data (x, y, z) are segmented into training and testing sets using the Look_Back and Forward_Length parameters, generating input-label pairs. A GRU-based deep learning model then processes each time sequence, and its outputs are used for near-real-time prediction of future UAV positions $(\hat{x}, \hat{y}, \hat{z})$.

Figure 1 illustrates how each step—namely, data and labeling, GRU modeling, real-time tuning, and evaluation—is integrated to produce an accurate and efficient UAV trajectory prediction solution. Subsequent sections detail the individual components in greater depth, including model architecture, hyperparameter ranges, and evaluation metrics.

2.2. Dataset Description and Pre-Processing

We employed the Zurich Urban Micro Aerial Vehicle (UMAV) Dataset [25], provided by the University of Zurich, to develop and evaluate our UAV trajectory prediction methodology. This dataset was collected using a Fotokite MAV (Perspective Robotics AG, Zurich, Switzerland), a tethered micro aerial vehicle designed for stable operation in urban environments.

The Fotokite MAV features a tethered power system, enabling continuous flight of up to 45 min, which is particularly advantageous for urban deployments where GPS signals may be unreliable. The UAV is controlled via a smartphone interface, allowing for precise

adjustments to yaw (rotation) and camera pitch angles. Its tethered design ensures safe and legally compliant operation in densely populated urban areas, where traditional UAVs may face regulatory restrictions.

Figure 2 shows an image of the Fotokite MAV used for data collection in the dataset.



Figure 2. Fotokite MAV used for data collection in the Zurich Urban Micro Aerial Vehicle (UMAV) Dataset. This tethered UAV is equipped with a GoPro Hero 4 camera and supports stable low-altitude flight (5–15 m) with continuous power supply for up to 45 min. Image source: Fotokite (www.fotokite.com), accessed on 5 February 2025.

The recorded parameters include time, latitude, longitude, altitude, speed, and heading, each sampled approximately every 0.03 s for a total flight duration of around 45 min. The raw log contains 81,169 data points, reflecting continual movement throughout multiple areas of a densely populated urban environment.

To conform to prevalent sampling rates (e.g., 0.1 s) in contemporary UAV trajectory prediction research (see Section 1.3), we downsampled the data from 0.03 s intervals to 0.1 s intervals, thus decreasing the dataset from 81,169 to 27,112 data points, achieving a compromise between temporal resolution and computational feasibility. Furthermore, to focus on core positional variables, we extracted only latitude, longitude, and altitude for training, discarding additional flight parameters (e.g., speed, heading). Preliminary analyses showed that these positional attributes alone sufficiently captured overall trajectory trends, particularly under the short-range maneuvering context central to this study.

For pre-processing, we evaluated various feature scaling methods frequently employed in deep learning and time-series forecasting. These encompassed StandardScaler [26], which subtracts the mean and divides by the standard deviation to produce a distribution with essentially zero mean and unit variance, and RobustScaler [27], which uses the median and interquartile range to alleviate the impacts of outliers. Although these methods effectively handle varied data distributions, extreme values can still unduly influence the scaling in certain scenarios.

In contrast, the Min–Max Scaler [28] rescales each feature to a fixed interval (commonly [0, 1]) by subtracting the minimum value and dividing it by the feature range (max–min). This method maintains the original distribution’s overall form and guarantees that all values are confined inside a specified numeric range. Despite its sensitivity to outliers, our dataset displayed a limited number of anomalies and a somewhat uniform distribution in latitude, longitude, and altitude, rendering Min–Max scaling a pragmatic option for normalizing feature magnitudes without compromising temporal correlations.

Thus, we utilized the Min–Max Scaler on latitude, longitude, and altitude values before model training, ensuring that each coordinate dimension was confined to a consistent range of [0, 1]. This normalization enabled steady convergence during training and streamlined hyperparameter optimization for the deep learning network. Ultimately, we divided the resampled dataset—consisting of 27,112 data points at 0.1 s intervals—into training (80%) and testing (20%) subsets, establishing the foundation for all ensuing experiments in this study.

2.3. Trajectory Labeling Method

To systematically train and evaluate our deep learning model for UAV trajectory prediction, we adopted a labeling scheme that divides flight data into overlapping input–output windows based on two parameters: Look_Back and Forward_Length (Figure 3). By tailoring the size of each historical context (Look_Back = L) and future forecast (Forward_Length = F), the method accommodates both short-term corrections and extended planning in low-altitude urban environments.

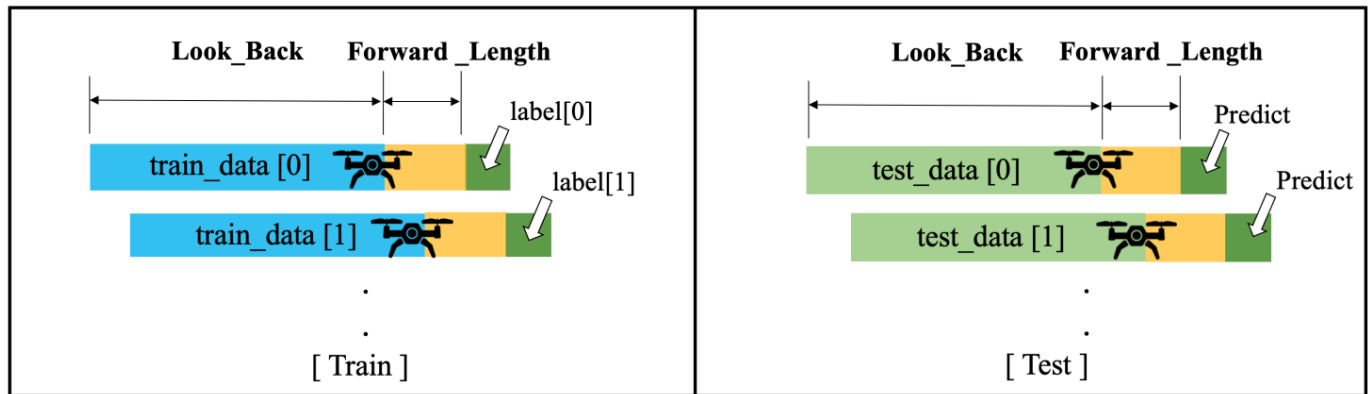


Figure 3. Labeling scheme for constructing the UAV trajectory prediction model, based on Look_Back and Forward_Length. Each flight log is segmented into overlapping windows: a Look_Back portion provides recent history, and the single future point at $t + F$ serves as the label (or predicted coordinate). On the training side, these windows produce labeled samples, while the testing side enables inference of unseen trajectories.

Look_Back (Equation (1)) determines how many consecutive data points (latitude, longitude, altitude) immediately preceding a time point t that the model observes. For instance, if Look_Back = L , then the window spans $(t - L + 1) \dots t$. Providing the network with a suitable Look_Back enables it to learn localized patterns in low-altitude, obstacle-rich settings, where abrupt heading changes and speed shifts are commonplace. Conversely, an overly large Look_Back may dilute the most relevant recent dynamics or inflate the computational load.

Forward_Length indicates how far in the future the model predicts a single coordinate, defined at $t + F$. A smaller F focuses on near-term collision avoidance and immediate corrections, whereas a larger F allows the model to see further into the future, enabling more proactive path reconfiguration—such as circumventing large obstacles well in advance, rerouting around urban corridors, or scheduling complex maneuvers long before the UAV actually encounters those conditions.

Let $\{x_i\}_{i=1}^N$ denote the time-sequenced UAV states (latitude, longitude, altitude) across a flight log of length N . For each valid index t (i.e., $t \leq N - F$), we define:

$$Input_t = (x_{t-L+1}, x_{t-L+2}, \dots, x_t). \quad (1)$$

$$Output_t = x_{t+F}. \quad (2)$$

If $t - L + 1 < 1$ or $t + F > N$, we exclude that partial segment.

Once we established Look_Back and Forward_Length, we labeled the dataset as follows. After splitting raw flight logs into training and testing sets, each valid index t in the training subset denotes an input segment of Look_Back points from $(t - L + 1)$ to t , and a single future output $t + F$ (Equation (2)). This procedure yields multiple overlapping input–label pairs along each UAV flight route. If $t + F$ extends beyond the last recorded data index, the partial segment is excluded to maintain consistent input–

output dimensions. An identical labeling step was applied in the testing subset, ensuring that the model—already trained on windows of length L —would produce forecasts for the single coordinate at $t + F$ in unseen trajectories. The resulting predictions were then compared to the ground-truth coordinates (e.g., using RMSE) to assess how effectively the model had learned and generalized for UAV trajectory prediction under diverse operational conditions.

By segmenting the data around these two parameters, the proposed labeling framework supplies the model with a well-defined recent history of flight states and an explicit future range to predict. This design allows the GRU-based architecture to accommodate the short-range, dynamic maneuvers characteristic of low-altitude UAV operations while still handling extended lookahead requirements. The following section outlines how these labeled sequences are fed into our deep learning network and discusses the training procedure, hyperparameter settings, and real-time feasibility analyses.

2.4. Deep Learning Model Architectures for UAV Trajectory Prediction

A central aim of this work is to identify a high-performance deep learning model capable of real-time UAV trajectory prediction under urban flight conditions, optimizing both accuracy and computational efficiency to meet the stringent demands of low-altitude operations. To this end, we developed and systematically compared five RNN variants—a vanilla RNN, LSTM, Bi-LSTM, GRU, and Bi-GRU. Each architecture processes labeled sequences (Section 2.3) to forecast future positions in latitude, longitude, and altitude. Transformer-based models (e.g., BERT) were initially considered but were excluded after preliminary tests revealed excessive parameter counts and higher inference latencies, making them incapable of satisfying the near-real-time requirements essential for low-altitude UAV operations.

2.4.1. Detailed Model Architectures

The vanilla RNN exemplifies the most basic recurrent architecture. Our approach consists of two successively stacked RNN layers, each containing 64 hidden units with *tanh* activation. The concealed state from the second RNN layer is transmitted to a fully connected layer with three output neurons, which directly forecast latitude, longitude, and altitude. This model, albeit possessing a limited number of parameters and serving as a baseline for sequence modeling, is susceptible to vanishing or bursting gradients in longer sequences, hence constraining its ability to capture prolonged flight patterns.

To mitigate the gradient problems associated with basic RNNs, the LSTM architecture has input, forget, and output gates, as well as a cell state that transmits information across time steps. Similar to the vanilla RNN, our LSTM solution comprises two stacked layers, each containing 64 hidden units with *tanh* activation, succeeded by a fully linked layer with three output neurons. LSTMs excel at learning medium- to long-term dependencies, rendering them appropriate for UAV movements that exceed short Look_Back windows. Nevertheless, the elevated parameter count of LSTM layers can impede training speed and augment the inference latency, potentially compromising its real-time performance.

The Bi-LSTM architecture enhances the functionality of the LSTM by analyzing input sequences in both directions, hence offering a more thorough comprehension of local flight circumstances. Each bidirectional layer consists of 64 hidden units in each direction (totaling 128 per layer), succeeded by a fully linked layer with three output neurons. This design can improve trajectory predictions in intricate situations by utilizing context from both preceding and subsequent locations within the Look_Back window. Nonetheless, the bidirectional operation doubles the computational burden, potentially affecting its appropriateness for stringent real-time applications.

The GRU architecture streamlines the LSTM design by substituting its three gates with update and reset gates, omitting the distinct cell state. This optimized design minimizes the number of parameters and frequently meets or surpasses the performance of LSTM in specific time-series tasks. Our GRU model employs two stacked layers, each with 64 hidden units and tanh activation, followed by a fully connected layer with three output neurons. GRUs are particularly advantageous for near-real-time UAV applications, in which a minimal computational overhead is critical due to sub-0.1 s inference intervals.

The Bi-GRU architecture ultimately integrates bidirectional processing with the GRU's streamlined gating mechanism. This model employs two bidirectional GRU layers, each with 64 hidden units per direction, to forecast latitude, longitude, and altitude via a fully linked layer with three output neurons. The Bi-GRU architecture, albeit marginally more computationally intensive than a unidirectional GRU, effectively captures intricate temporal patterns, making it advantageous for difficult low-altitude flight situations.

2.4.2. Hyperparameters and Experiment Setup

Across all five RNN architectures, we reshaped each sample into a (batch_size, Look_Back, 3) format, where the three features represent latitude, longitude, and altitude, and then passed the final hidden states through a dense layer with three neurons for position prediction. We used tanh activation functions in the recurrent layers and applied a linear projection in the output layer. Each network was optimized via mean squared error (MSE) loss and the Adam optimizer with an initial learning rate of 1×10^{-3} . Training proceeded for a maximum of 500 epochs, with an early stopping mechanism (patience set to 15 epochs) reverting to the best-performing epoch if the validation loss ceased improving. A consistent batch size of 70 was employed, and Glorot (Xavier) initialization was applied to both recurrent and dense layers. The models were largely trained on an NVIDIA RTX A6000 GPU (48 GB VRAM), providing efficient processing and uniform evaluation of all five architectures. The outcomes of these studies, encompassing precise accuracy measurements, data partitioning methodologies, and analyses of computational efficiency, are thoroughly examined in Section 2.5.

2.5. Metrics for Model Evaluation and Performance Analysis

We utilized various assessment criteria and experimental configurations to thoroughly assess the predicted efficacy and real-time viability of the suggested models. These evaluations aimed to test the models' overall accuracy, robustness across different prediction horizons, and alignment with real-time needs. This section delineates the measurements employed, the experimental configurations, and the visualization techniques utilized to analyze and present the data efficiently.

All assessments were performed utilizing a dataset comprised of 27,112 samples. The dataset was partitioned into training and testing subsets in an 80:20 ratio, yielding 21,690 samples for training and 5422 samples for testing. Models were trained exclusively on the training data and evaluated on the test data to ensure an unbiased assessment of their performance. The training and testing datasets were rigorously checked to ensure no overlap or duplication, allowing for an accurate evaluation of the models' generalization capabilities. This careful preparation ensured that the results reflected the true predictive performance of the models.

2.5.1. Evaluation Metrics: RMSE and MAE

The model performance was assessed using two established regression metrics—RMSE and MAE—which collectively offer a thorough evaluation of prediction accuracy and dependability. RMSE quantifies the square root of the mean of the squared discrepancies between expected and actual values. By emphasizing larger errors due to its squared error

term, RMSE is particularly sensitive to outliers, making it a suitable choice for scenarios where minimizing large deviations is critical. A lower RMSE value reflects higher predictive accuracy, indicating the model's ability to closely approximate the actual trajectory. The RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (3)$$

where y_i denotes the ground-truth value, $\hat{y}_i$ is the predicted value, and N is the total number of data points.

Conversely, MAE computes the mean of the absolute discrepancies between projected and actual values, providing a clear interpretation of prediction accuracy. In contrast to RMSE, MAE provides uniform weights to all errors, regardless of their size, hence offering an additional perspective on model efficacy. This equilibrium enables the MAE to represent consistent precision in forecasts without unduly prioritizing outliers. The MAE is defined as:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (4)$$

where y_i denotes the ground-truth value, $\hat{y}_i$ is the predicted value, and N is the total number of data points.

The RMSE and MAE were calculated for each model across several prediction situations in order to ensure a comprehensive assessment of accuracy. By employing these two criteria collectively, we guaranteed that the assessment encompassed both sensitivity to outliers and overall error consistency, facilitating a comprehensive review of each model's predictive strengths and weaknesses.

2.5.2. Tuning Look_Back, Extending Forward_Length, and Real-Time Feasibility

The first step in evaluating model performance was to determine the appropriate Look_Back value for each of the deep learning models. The Look_Back value is the length of the historical data used as an input for making the prediction, which is a very important factor in identifying the temporal patterns. To find the appropriate Look_Back value for each of the architectures, we performed an experiment with different input window sizes. This process helped to determine the Look_Back values that would give the best possible performance in the models without having to pay the cost of large computational power for a given prediction accuracy.

After setting the right Look_Back value for each model, we employed these configurations to make predictions for a longer time period. The Forward_Length, which denotes the period in which forward predictions are made, was incremented in steps to see how well the models performed when the time to predict the UAV position was increased. This way of examining each model's fidelity to choose the appropriate Look_Back value for each model and its ability to make adequate predictions for the next time step is important for enabling real-world applications in low-altitude urban environments.

In addition to assessing the prediction accuracy under different Look_Back and Forward_Length settings, we also checked how fast our deep learning models could generate predictions. We also checked whether the models could make the predictions within the typical time frame of the UAV data (i.e., 0.1 s) while remaining computationally efficient. By measuring the central processing unit (CPU) and graphics processing unit (GPU) resource utilization during inference, we attempted to logically justify the model's real-time capability without incurring high computational costs. This assessment guarantees that the proposed model is not only highly accurate but also viable for implementation in real-world UAV systems.

The analysis of Look_Back optimization, Forward_Length experiments, and real-time feasibility of the models offered insights into the temporal prediction properties of the models. By starting with the determination of the best Look_Back configuration, the Forward_Length evaluations were performed in the best possible condition of each model. This systematic evaluation not only supported the significance of fine-tuning of temporal input parameters but also indicated how well the models suit both short- and long-term predictions. This two-step process of identifying the optimal Look_Back values and then increasing Forward_Length provides a good approach for understanding the capabilities of models in the context of real-world UAV trajectory prediction tasks.

2.5.3. Visualization of Prediction Performance

Trajectory visualizations were utilized alongside numerical indicators (i.e., RMSE and MAE) to enhance the intuitive comprehension of model performance. This methodology enabled a direct comparison between the real UAV trajectories and the model predictions. By superimposing the predicted trajectories onto actual flight tracks, variations and trends can be visually discerned, emphasizing places where the models excelled or faltered.

Visualization provided numerous benefits. Initially, it enabled a transparent and prompt comparison among models, facilitating the evaluation of predictive accuracy instantaneously. Secondly, it facilitated the identification of trends in error accumulation that may not be readily discernible using numerical measures alone. Visualizations indicated distinct flight patterns, including abrupt turns and dramatic height variations, while particular models exhibited greater deviations from the actual track. Ultimately, these visualizations provided insights into the behavioral consistency of the models, demonstrating their capacity to navigate intricate trajectories in real-world contexts.

Besides facilitating assessment, trajectory visualizations were crucial in illustrating the practical applicability of the models. This enables stakeholders, including UAV operators and researchers, to grasp such data more intuitively and identify instances necessitating further refinement of the models. Through these comparisons, we confirmed the numerical performance measures and showed the practical significance of the suggested methods in dynamic UAV trajectory prediction tasks.

2.5.4. Performance Evaluation Summary

The evaluation approach involved thoroughly assessing model performance using quantitative measures, temporal input optimization, expanded prediction horizons, and real-time feasibility tests. The RMSE and MAE offered quantitative insights into predictive accuracy, whereas trajectory visualizations illuminated behavioral patterns and practical significance. The methodical adjustment of Look_Back and Forward_Length further illustrated the models' resilience across different temporal scales. Ultimately, real-time feasibility analysis validated the operational appropriateness of the suggested models, achieving a balance between accuracy and processing economy for actual UAV trajectory prediction tasks.

3. Results

3.1. Optimal Input Length and Identification of the Most Suitable Deep Learning Model for UAV Trajectory Prediction

This work systematically adjusted the Look_Back parameter, which specifies the duration of previous data utilized for prediction, across all five deep learning models: vanilla RNN, LSTM, Bi-LSTM, GRU, and Bi-GRU. We performed studies to determine the ideal Look_Back value for each model, testing values from 10 to 70 in increments of 10. The performance of the models was assessed using the RMSE and MAE measures across three

predictive dimensions: latitude, longitude, and altitude. These measures were selected due to their capacity to deliver a comprehensive evaluation of predictive accuracy and error consistency.

The optimization outcomes for the GRU model, as shown in Table 2, indicated that it attained optimal performance with a Look_Back value of 50. In this scenario, the GRU model achieved the lowest average RMSE of 0.0621, with a prediction time of 0.0334 s, demonstrating its proficiency in capturing temporal dependencies while ensuring computational economy. Nonetheless, extending the Look_Back beyond this threshold resulted in declining results, as evidenced by marginal increases in RMSE and computational burden, suggesting the likelihood of over-fitting or the incorporation of superfluous historical data.

Table 2. Performance of the GRU model across different Look_Back values, evaluated using RMSE and MAE metrics for latitude (Lat), longitude (Lon), and altitude (Alt). The average RMSE across all dimensions and the prediction time in seconds for each Look_Back configuration are also included. The optimal Look_Back value of 50 is highlighted in bold, achieving the lowest average RMSE and competitive prediction time.

Look_Back	RMSE			MAE			Average RMSE	Prediction Time (s)
	Lat	Lon	Alt	Lat	Lon	Alt		
10	0.000011	0.000009	0.2166	0.000009	0.000007	0.1353	0.0722	0.0323
20	0.000012	0.000005	0.2021	0.000011	0.000004	0.1058	0.0673	0.0319
30	0.000007	0.000007	0.185643	0.000005	0.000006	0.0798	0.0618	0.0341
40	0.000009	0.000010	0.1890	0.000007	0.000008	0.0852	0.0630	0.0334
50	0.000005	0.000002	0.1864	0.000004	0.000002	0.0795	0.0621	0.0334
60	0.000008	0.000005	0.188091	0.000007	0.000004	0.0798	0.0627	0.0346
70	0.000016	0.000009	0.2042	0.000014	0.000008	0.1110	0.0680	0.0341

Comparable tendencies were noted for the other models, with performance reaching its zenith at particular Look_Back values contingent upon their architectural attributes. The ideal Look_Back values for these models were as follows: the vanilla RNN excelled at a Look_Back of 40, while the LSTM model reached peak performance with a Look_Back of 60. The Bi-LSTM architecture demonstrated optimal accuracy and efficiency with a Look_Back of 70, indicating its capacity to analyze extended input sequences in both directions. The Bi-GRU model demonstrated superior performance with a diminished Look_Back of 30, suggesting that its efficient gating mechanism and bidirectional structure operated most effectively with smaller input windows.

To thoroughly evaluate the overall performance of these models under their best Look_Back configurations, we examined their accuracy and computational efficiency, as outlined in Table 3. The GRU model consistently outperformed the other designs, achieving the lowest average RMSE and MAE across all predictive metrics. Furthermore, its prediction time of 0.0334 s demonstrated its remarkable computational efficiency, underscoring its suitability for real-time UAV trajectory forecasting applications. The vanilla RNN and Bi-GRU models demonstrated competitive outcomes; however, their marginally elevated RMSE and MAE values, along with extended prediction durations, positioned them as inferior to the GRU in overall performance. Likewise, the LSTM and Bi-LSTM models demonstrated proficiency in identifying intricate temporal patterns; nevertheless, their architectural complexity resulted in elevated computing expenses, rendering them sub-optimal for real-time UAV applications.

These results underscore the necessity of customizing the Look_Back parameter to the design of each model. The variations in optimal Look_Back values can be ascribed to the distinct temporal processing methods inherent to each design, including the gating mechanisms in LSTM and GRU, as well as the bidirectional functionality in Bi-LSTM and Bi-GRU. By determining the ideal Look_Back value for each model, we guaranteed that the

following analyses were performed under the most advantageous conditions, facilitating equitable and significant comparisons of their prediction performance. Table 2 illustrates the Look_Back optimization outcomes of the GRU model, while Table 3 offers a comprehensive comparison of all models in their optimal configurations. The thorough testing highlighted the essential function of Look_Back tuning in improving model performance and reconciling accuracy with computing efficiency, eventually determining the GRU as the superior model for UAV trajectory prediction.

Table 3. Performance comparison of five deep learning models under their optimal Look_Back configurations. Metrics include RMSE and MAE for latitude (Lat), longitude (Lon), and altitude (Alt) predictions, as well as the average RMSE across all dimensions and prediction time (in seconds). The GRU model achieved the best overall performance with the lowest average RMSE and MAE, as well as competitive prediction time, making it the most suitable model for UAV trajectory prediction.

Model	Look_back	RMSE			MAE			Average RMSE	Prediction Time (s)
		Lat	Lon	Alt	Lat	Lon	Alt		
Bi-GRU	30	0.000006	0.000003	0.1920	0.000005	0.000002	0.0941	0.0640	0.0343
GRU	50	0.000005	0.000002	0.1864	0.000004	0.000002	0.0795	0.0621	0.0334
Bi-LSTM	70	0.000003	0.000004	0.2091	0.000002	0.000003	0.1110	0.0697	0.0347
LSTM	60	0.000005	0.000003	0.1989	0.000004	0.000002	0.0943	0.0663	0.0345
RNN	40	0.000003	0.000004	0.1908	0.000002	0.000003	0.0803	0.0636	0.0355

To enhance the validation of the models' predictive performance, we graphed the predicted trajectories in conjunction with the actual trajectory as a 3D map. Figure 4a illustrates that all five deep learning models were assessed on a test trajectory marked by frequent directional shifts and significant variability, which are common issues in UAV flight trajectories. Notwithstanding the intrinsic complexity of the trajectory, the GRU model exhibited the most accurate correspondence with the actual trajectory, surpassing alternative designs in terms of both precision and stability.

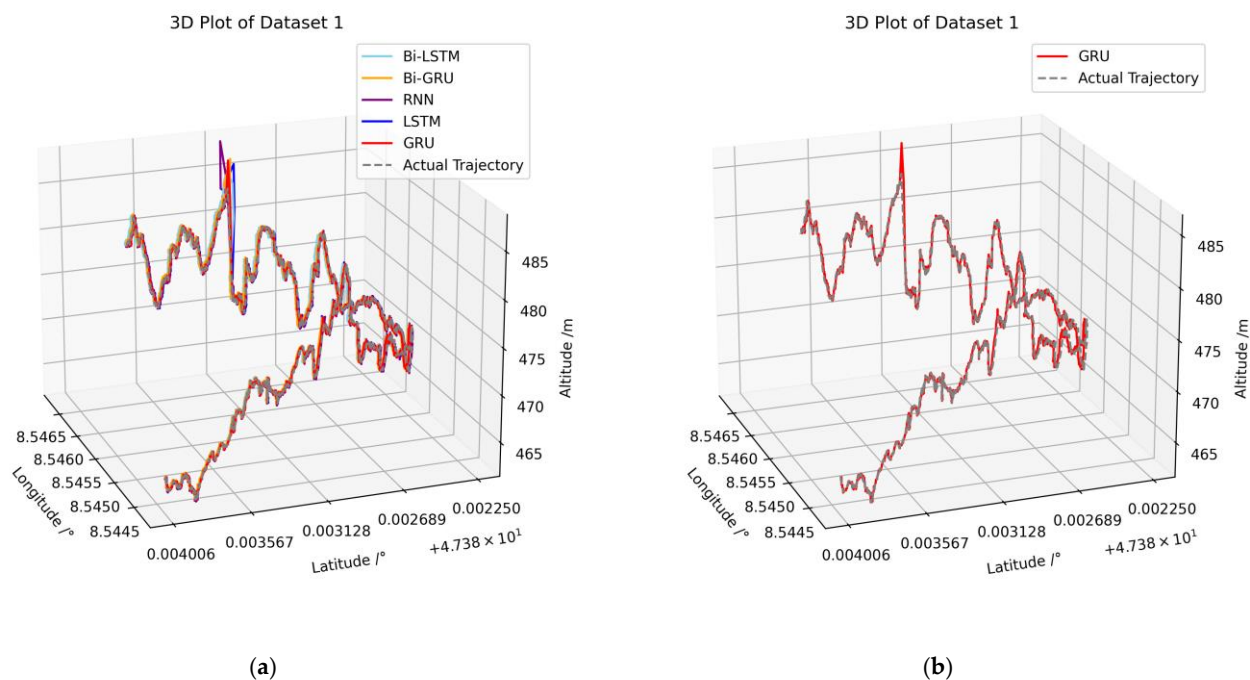


Figure 4. (a) Comparison of trajectories predicted with five deep learning models (Bi-LSTM, Bi-GRU, RNN, LSTM, and GRU) and the actual trajectory in a 3D space. (b) Comparison of the GRU model's predicted trajectory with the actual trajectory, illustrating its superior alignment and accuracy with respect to challenging UAV flight paths characterized by frequent direction changes.

Figure 4b exclusively illustrates the GRU model's prediction in relation to the actual trajectory. This graphic illustrates the GRU model's proficiency in precisely capturing both abrupt shifts and gradual transitions, even within a highly dynamic context. The results highlight the model's resilience and its capacity to manage real-world UAV trajectory prediction jobs with negligible error. The visuals, along with the numerical measurements, highlight that the GRU model offers computational efficiency while also excelling in accurately forecasting UAV trajectories in intricate and demanding circumstances.

Unlike many deep learning architectures with multiple gating mechanisms, the GRU model (Figure 5) employs only an update gate and a reset gate. This streamlined design facilitates rapid adaptation to the frequent heading changes and complex dynamics common in UAV flights, offering a clear rationale for why GRU-based models often achieve best-in-class performance in UAV trajectory prediction. The equations for the GRU gates are as follows:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]). \quad (5)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]). \quad (6)$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t * h_{t-1}, x_t]). \quad (7)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t. \quad (8)$$

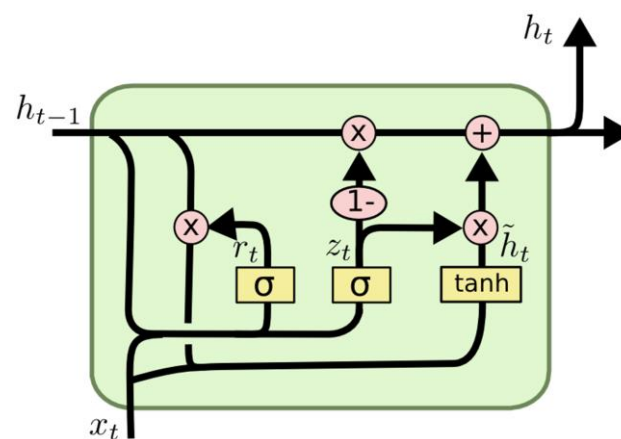


Figure 5. Internal structure of a Gated Recurrent Unit (GRU) cell. The update gate (z_t) determines how much past information (h_{t-1}) is kept, while the reset gate (r_t) selectively resets the old state. A tanh function then forms a candidate hidden state $\tilde{h}_t$, and the final state h_t blends $\tilde{h}_t$ with h_{t-1} . This gating mechanism enables rapid adaptation to sudden flight changes, making GRUs well-suited for UAV trajectory prediction.

As seen in Equation (5), the update gate (z_t) governs how much historical information to retain, while the reset gate (r_t) in Equation (6) selectively resets the old state before forming a candidate hidden state $\tilde{h}_t$, (Equation (7)). The final state (h_t) is then computed via a weighted combination of (h_{t-1}) and $\tilde{h}_t$ (Equation (8)).

One more notable feature, as seen in the formulation, is that no activation function is applied to the final output. Combined with the GRU's unique characteristics—utilizing only two gates and integrating the cell state and hidden state—these factors enable the model to achieve the best performance in predicting the challenging trajectories of UAVs.

3.2. Real-Time Feasibility of Trajectory Prediction for UAV

We previously conducted experiments on a system with an Intel® Xeon® Gold 6346 CPU (3.10 GHz, 128 GB RAM) and an NVIDIA RTX A6000 GPU to assess the performance of our deep learning model for trajectory prediction in UAV scenarios. Table 3 illustrates

that the GRU model attained an average inference time of 0.0334 s per sample, thus meeting the target criterion of 0.1 s for real-time prediction. The 0.1 s threshold denotes the data update cycle typically utilized in UAV systems, wherein sensor data are refreshed and processed every 100 milliseconds to guarantee precise and prompt trajectory modifications.

Furthermore, we concentrated on determining the inference time and performed tests to assess the degree of CPU and GPU utilization during model inference. The supplementary experiments sought to empirically confirm the real-time viability of our model and elucidate its computational requirements. These measurements act as a surrogate for comprehending the integration capability of the model within actual UAV systems. We aimed to evaluate the resource utilization of our model, in order to determine whether it incurs substantial computational overhead or allows adequate capacity for other critical activities, such as sensor data management and flight control.

In the inference phase, the average GPU utilization was 77.78%, which implies that the computationally expensive tasks were effectively shifted to the parallel computing units of the GPU. In the meantime, the CPU utilization was as low as 7.56%, which reveals the light workload of the CPU. This nearly zero CPU usage is particularly useful in a UAV system, which allows other critical functions, such as sensor data processing, communication, or flight control, to run without a significant loss of performance.

The major limitation is that the achieved results were obtained on high-performance hardware (i.e., the RTX A6000 GPU). This investigation mainly tested the real-time capability of our model and did not address the issue of identifying the optimal hardware configuration. When the GPU resources are limited or some of them are not available, it may be possible to achieve similar performance by applying optimization techniques such as model pruning, quantization, knowledge distillation, or increasing batch sizes in practical applications. The obtained inference time of 0.0334 s demonstrates that real-time performance is achievable on mid-range GPUs or other specialized computing devices with only slightly increased latencies.

These findings illustrate the potential for integrating our trajectory prediction model into a UAV system. The sub-0.1 s inference time and the almost negligible CPU usage are advantageous for applying this method for real-time trajectory prediction in dynamic and complex UAV environments.

3.3. Extended Prediction Horizons Analysis

To assess the GRU model's capacity to forecast trajectories across prolonged timeframes, we executed tests by incrementally augmenting the Forward_Length in accordance with the optimal Look_Back value of 50, as determined in the preceding sections. Table 4 displays the experimental findings, with Forward_Length ranging from 5 (0.5 s) to 40 (4 s). As anticipated, an increase in Forward_Length led to elevated errors, given that the prediction targets extend longer into the future, complicating trajectory estimation.

The objective of this experiment was to observe the increase in error with distance and to test the model's capacity to sustain robust predictive performance across extended trajectory horizons. The findings indicated that the GRU model exhibits consistent performance across different Forward_Length values, maintaining an average RMSE below 1 until the Forward_Length approaches 40 (4 s). Beyond this threshold, the inaccuracy escalates considerably, prompting us to establish 40 as the maximum Forward_Length value in this study.

These findings suggest that the GRU model can effectively predict trajectories up to 4 s into the future—a critical capability, given the rapid velocity changes typical of UAVs. Even at this extended Forward_Length, the average RMSE of 0.7198 and the prediction time of 0.0364 s indicate that the model remains both accurate and computationally efficient. These results confirm the model's relevance in contexts necessitating short- and medium-term trajectory forecasts.

Table 4. Performance of the GRU model across different Forward_Length values, ranging from 5 (0.5 s) to 40 (4 s), based on the optimal Look_Back value of 50. Metrics include RMSE and MAE for latitude (Lat), longitude (Lon), and altitude (Alt), as well as the average RMSE across all dimensions and prediction time (in seconds). The results demonstrate the GRU model's ability to maintain strong predictive performance and computational efficiency for trajectory predictions up to 4 s into the future.

Forward_Length (s)	RMSE			MAE			Average RMSE	Prediction Time (s)
	Lat	Lon	Alt	Lat	Lon	Alt		
5 (0.5)	0.000022	0.000026	0.5384	0.000020	0.000025	0.2889	0.1794	0.0331
10 (1)	0.000016	0.000009	0.8393	0.000013	0.000007	0.5025	0.2798	0.0332
15 (1.5)	0.000019	0.000017	1.1591	0.000016	0.000014	0.7694	0.3863	0.0330
20 (2)	0.000016	0.000020	1.6509	0.000012	0.000016	1.1737	0.5503	0.0330
30 (3)	0.000054	0.000094	2.5052	0.000043	0.000078	1.8741	0.8351	0.0353
40 (4)	0.000088	0.000077	2.1593	0.000080	0.000057	1.5707	0.7198	0.0364

A significant observation is the comparatively elevated error rates in altitude (Alt) predictions, relative to latitude (Lat) and longitude (Lon). This behavior can be ascribed to the intrinsic complexity of altitude dynamics in UAV environments, where rapid vertical movements and turbulence occur more frequently than horizontal shifts. These problems highlight the necessity of enhancing altitude prediction models, possibly through integrating supplementary variables such as wind speed, air pressure, or turbulence data to augment performance in this area.

The results detailed in Table 4 demonstrate the efficacy of the GRU model in managing lengthy trajectory prediction tasks, underscoring its robustness and applicability for UAV applications. The uniform performance across different Forward_Length values highlights its capability to deliver dependable trajectory predictions, even in swiftly evolving urban flight conditions.

4. Discussion

4.1. Comparative Analysis of UAV and Aircraft Trajectory Prediction

This work commenced by examining the existing literature on trajectory prediction for both aircraft and UAV systems. The prediction of aircraft trajectories has been thoroughly investigated, due to the accessibility of substantial open datasets and the prevalent use of deep learning in the aviation sector. Through an analysis of these studies, we noticed conceptual parallels between aircraft and UAV trajectory prediction, which informed the creation of our model.

Nonetheless, aircraft and UAV trajectory prediction tasks exhibit considerable divergence in complexity and operational environment. Aircraft often operate at elevated altitudes with little directional alterations, leading to trajectories that are reasonably stable and predictable. This stability facilitates more substantial increases in Forward_Length during predictive experiments without markedly affecting performance. Conversely, UAV systems typically function in low-altitude, urban settings marked by rapid directional shifts, turbulence, and dynamic obstructions. These challenges render substantial Forward_Length increments more arduous to attain without incurring forecast errors. Notwithstanding these intrinsic complexities, our model showed robust efficacy in forecasting UAV trajectories over diverse Forward_Length values. Although numerical errors escalated progressively with increasing Forward_Length, the lack of sudden spikes demonstrated the model's resilience in managing the majority of trajectories. Visualizations indicated that a limited number of outliers appeared at greater Forward_Length values, illustrating the swift directional shifts characteristic of UAV systems. To highlight the most reliable outcomes, visualizations in this investigation were performed at Forward_Length = 0, when the model exhibited the closest alignment with the real trajectories.

This study distinguishes itself from most prior UAV trajectory prediction research by integrating a direct comparison with an external model, thus establishing a new norm for rigor and transparency, rather than solely concentrating on internal model evaluation. To emphasize the advantages of our proposed model, we performed a comparative analysis by training and evaluating FlightBERT++—an advanced aircraft trajectory prediction model—on the same UAV dataset. FlightBERT++ enhances the original FlightBERT architecture by employing a BERT-based framework tailored for high-altitude aircraft trajectories. This model integrates a trajectory encoder to capture spatial and temporal characteristics, a horizon embedding module for multi-step forecasts, and a differential-prompted decoder to improve accuracy in directional alterations.

The comparative results, as outlined in Table 5, illustrate the enhanced performance of our proposed GRU-based model when compared to FlightBERT++ on the same UAV dataset. In addition to the accuracy improvements, our GRU-based model also offers a significant advantage in terms of computational efficiency. While FlightBERT++ benefits from its transformer-based architecture, it requires substantially higher memory and processing power, making real-time deployment more challenging. The simplicity of GRU allows for faster inference times and lower energy consumption, which are crucial factors in UAV applications where on-board computational resources are limited.

Table 5. Performance comparison of the GRU model and FlightBERT++ on RMSE across prediction dimensions (latitude, longitude, altitude) using the UAV dataset. Our model demonstrates superior accuracy, highlighting its applicability for real-time UAV trajectory prediction.

Model	RMSE		
	Lat	Lon	Alt
Ours	0.000005	0.000002	0.1864
FlightBERT++	0.0053	0.0240	5.5894

The comparison was performed at Forward_Length = 0, where both models attained their peak performance. Notably, we observed that the FlightBERT++ model converged very quickly during training, with its loss nearly reaching zero by epoch 4, at which point its best model checkpoint was saved. Despite this rapid convergence, FlightBERT++ (latitude: 0.0053, longitude: 0.0240, altitude: 5.5894) underperformed our GRU-based model (latitude: 0.000005, longitude: 0.000002, altitude: 0.1864). This disparity highlights our model's ability to adapt to the swift directional changes and complex dynamics typical of UAV datasets, in contrast to FlightBERT++, which is tailored more for stable aircraft trajectories. Moreover, the lower RMSE values in all dimensions demonstrate that our GRU-based model not only addresses the inherent challenges of UAV trajectory prediction, such as abrupt heading shifts and intricate flight paths, but also exhibits improved generalization capacity without overfitting on the training data.

4.2. Generalization Performance Evaluation on Independent UAV Datasets

To evaluate the generalization performance of the proposed GRU-based trajectory prediction model, experiments were conducted on two independent UAV datasets. These datasets were chosen with the aim of evaluating the model's capacity to predict previously unencountered trajectories, thus offering a comprehensive assessment of its adaptability and reliability in practical applications.

The initial dataset—Synthetic UAV Flight Trajectories [29]—was created by RIOTU Lab and comprises synthetically generated UAV trajectories. This dataset contains temporal, latitude, longitude, and altitude data for UAVs adhering to specified trajectories. Data points were collected at intervals of approximately 0.1 s, resulting in a total flight duration of 25 s and comprising 251 data points per trajectory.

The second dataset—the Package Delivery Quadcopter Drone Dataset [30]—was supplied by Carnegie Mellon University and comprises actual quadcopter drone flights intended for package delivery operations. The flights utilized drones fitted with onboard sensors to capture data on time, latitude, longitude, altitude, velocity, and heading. Data were collected at intervals of 0.1 to 0.3 s, with each trajectory lasting approximately 7 min and comprising 2856 data points. This dataset presents considerable variability arising from diverse payloads, environmental conditions, and dynamic flight patterns, rendering it suitable for evaluating the model’s robustness in practical UAV operations.

The GRU-based model was evaluated on both datasets without supplementary training or fine-tuning, thereby simulating a realistic scenario in which the model is utilized to predict trajectories of previously unobserved UAVs. Predictions were assessed at different Forward_Length values, with increments of 5 steps, in order to analyze the model’s performance across short- and long-term prediction horizons. This assessment sought to evaluate the model’s capacity for effective generalization to new trajectories, independent of the characteristics of the underlying dataset.

The findings presented in Table 6 regarding the Synthetic UAV Flight Trajectories dataset indicate that the model consistently attained low RMSE and MAE values for shorter Forward_Length values. Despite the model’s lack of prior exposure to this dataset, it demonstrated high accuracy in predictions, especially for short-term horizons. As Forward_Length increased, its prediction errors also increased, indicating the inherent challenges associated with forecasting longer trajectories. The increase in error was gradual rather than abrupt, demonstrating the model’s robustness in managing extended predictions. To provide a clearer view of the model’s performance, visualization was conducted for Forward_Length = 0, as shown in Figure 6. This choice was made to highlight the scenario in which the model most closely followed the actual trajectory.

Table 6. Performance of the GRU-based model on the Synthetic UAV Flight Trajectories dataset across varying Forward_Length values. Metrics include RMSE and MAE for latitude, longitude, and altitude.

Forward_Length (s)	RMSE			MAE			Average RMSE	Prediction Time (s)
	Lat	Lon	Alt	Lat	Lon	Alt		
0 (0.1)	0.000012	0.000009	0.2853	0.000009	0.000006	0.2139	0.095107	0.0354
5 (0.5)	0.000034	0.000021	0.7202	0.000027	0.000013	0.6153	0.240085	0.0341
10 (1)	0.000059	0.000046	2.3676	0.000045	0.000029	1.8635	0.789235	0.0334
15 (1.5)	0.000089	0.000043	4.0401	0.000067	0.000032	3.0698	1.346744	0.0332
20 (2)	0.000131	0.000067	11.9086	0.000105	0.000048	7.8398	3.969599	0.0338

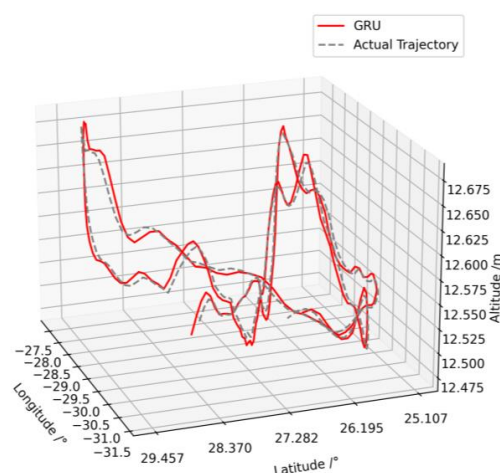


Figure 6. Predicted vs. actual trajectories on the Synthetic UAV Flight Trajectories dataset, consisting of 251 data points per trajectory. The visualization highlights our model’s ability to closely follow pre-defined paths with high accuracy under synthetic conditions.

On the Package Delivery Quadcopter Drone dataset, as shown in Table 7, the GRU-based model exhibited competitive performance despite the increased variability inherent to real-world flight data. The model maintained high accuracy for short-term predictions, with errors growing moderately as the Forward_Length value increased. The absence of abrupt error surges demonstrates the model's resilience to the dataset's complex dynamics and external factors such as environmental noise and payload variations. Figure 7, similar to the synthetic dataset, emphasizes the case where Forward_Length = 0, illustrating the model's capacity to adjust to real-world UAV scenarios while preserving consistency with the actual trajectory.

Table 7. Performance of the GRU-based model on the Package Delivery Quadcopter Drone dataset across varying Forward_Length values. Metrics include RMSE and MAE for latitude, longitude, and altitude.

Forward_Length (s)	RMSE			MAE			Average RMSE	Prediction Time (s)
	Lat	Lon	Alt	Lat	Lon	Alt		
0 (0.1)	0.111855	0.189168	0.0102	0.099947	0.169011	0.0079	0.10347	0.0437
5 (0.5)	0.466033	0.783567	0.0523	0.418435	0.696838	0.0391	0.433967	0.0440
10 (1)	0.773445	1.389192	0.0907	0.685822	1.240606	0.0725	0.751112	0.0412
15 (1.5)	1.104194	1.887841	0.0944	0.992838	1.703894	0.0742	1.028812	0.0420
20 (2)	1.448889	2.324619	0.1065	1.340334	2.112503	0.0865	1.293336	0.0428

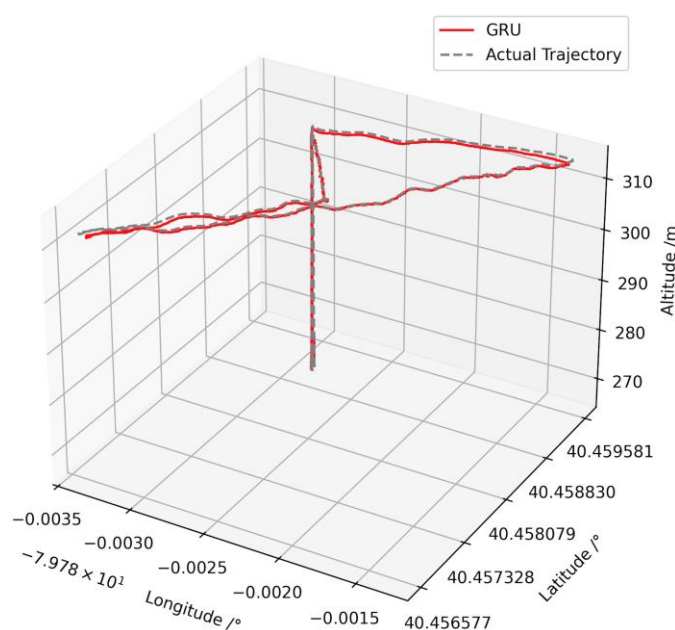


Figure 7. Predicted vs. actual trajectories on the Package Delivery Quadcopter Drone dataset, consisting of 2856 data points per trajectory. The visualization demonstrates our model's adaptability to dynamic real-world flight paths with varying payloads and environmental conditions.

The experimental results indicate that the proposed GRU-based model demonstrates robust generalization across both synthetic and real-world datasets. In testing on previously unseen UAV trajectory logs, the model often achieved equal or superior performance compared to best-known results from other UAV prediction approaches, strongly suggesting exceptional scalability. Prediction accuracy remained high on unfamiliar data, with a gradual increase in error at longer Forward_Length values. This outcome underscores the model's scalability for a wide range of UAV applications, from controlled research settings to dynamic operational contexts. Furthermore, visualizations at Forward_Length = 0 showed that the model can closely track actual flight trajectories under optimal conditions.

4.3. Potential Applications and Future Impact of Our Deep Learning Study

This study's findings offer substantial potential for the swiftly advancing domain of UAVs, which is expected to revolutionize urban transportation in the forthcoming decades. Precise trajectory prediction is essential for guaranteeing the safety, efficiency, and scalability of UAV operations, and this research tackles one of the sector's most formidable technological challenges. This project serves as a basis for expanding UAV systems in real-world applications by constructing a robust and adaptive trajectory prediction model.

A significant obstacle to the extensive implementation of UAVs is guaranteeing operational safety and reliability. This research directly tackles this crucial concern by offering a trajectory prediction model that can markedly enhance the safety and reliability of UAV systems. This research provides a credible method for precise trajectory prediction, hence aiding in the creation of UAVs as a trustworthy and widely accepted form of urban transportation.

The most immediate application of this research is the enhancement of UAV safety via real-time collision prediction and avoidance. With accurate trajectory predictions, UAVs can foresee probable accidents with other aerial vehicles or obstacles and dynamically modify their routes to reduce hazards. This capacity is crucial in crowded urban airspaces, where UAV density is anticipated to increase significantly with the proliferation of UAV usage. Real-time collision prediction, enabled by our proposed model, involves continuously forecasting the trajectories of multiple UAVs and identifying potential conflicts in advance. If a collision risk is detected, the system can issue immediate warnings or autonomously adjust flight paths to avoid accidents. This proactive approach enhances the reliability of UAV operations, particularly in dynamic and high-traffic environments, ensuring safer navigation and minimizing disruptions.

In addition to collision avoidance, the trajectory prediction technology established in this study can significantly contribute to counter-drone systems. With the rise in UAV utilization, the likelihood of illicit or harmful drone operations in prohibited zones has also increased. By precisely forecasting the trajectories of these drones, security systems can respond more efficiently, intercepting or eliminating threats prior to their escalation. This application illustrates the adaptability of trajectory prediction algorithms in tackling new security concerns as well as economic and logistical tasks.

The technique is equally beneficial for enhancing UAV operations in package delivery, medical logistics, and emergency response contexts. Precise trajectory forecasts enable operators to optimize energy usage, decrease delivery durations, and enhance overall mission efficacy, particularly in intricate circumstances characterized by variable weather conditions or constrained airspace. Medical drones transporting essential supplies to inaccessible regions can depend on reliable forecasts to guarantee prompt and precise delivery, potentially preserving lives during catastrophes.

This research significantly influences society and the economy by facilitating the adoption of UAVs. This study addresses critical issues relating to trajectory prediction, hence facilitating the integration of UAVs into extensive transportation networks. UAVs promise to alleviate ground traffic congestion, decrease greenhouse gas emissions, and enhance accessibility to vital commodities and services. The exhibited performance of the suggested model represents a concrete advancement toward achieving these advantages, highlighting its potential as a fundamental facilitator of the UAV ecosystem.

This research significantly contributes to UAVs by tackling essential issues in trajectory prediction. Its usefulness encompasses safety-critical operations, anti-drone systems, and mission optimization across many contexts. This study enhances the safety and trustworthiness of UAVs, providing the essential technology to realize the full potential of UAVs, thus facilitating their widespread adoption and transformational influence on urban mobility.

4.4. Contributions of This Study

The results of this study offer considerable potential for the swiftly advancing domain of UAVs. This study significantly contributes to UAV trajectory prediction by addressing key constraints in the existing literature and creating new benchmarks in terms of predictive performance, long-range forecasting, real-time practicality, and generalization capabilities.

1. Development of the Most Accurate UAV Trajectory Prediction Model

Through a comparative investigation of various deep learning architectures, we determined that the GRU-based model is the most suitable option for UAV trajectory prediction. We labeled the sequential trajectory data for training the deep learning network using Look_Back and Forward_Length, establishing a basis for identifying temporal trends in UAV trajectories. By utilizing this labeling methodology and refining parameters such as Look_Back and Forward_Length, we created a model that attained exceptional predictive performance. The suggested model achieved RMSE values of latitude: 0.000005, longitude: 0.000002, and altitude: 0.1864, resulting in an average RMSE of 0.0621, surpassing all compared UAV and aircraft trajectory prediction models.

Additionally, consistent with recent research, such as that of Palamas [5], who primarily utilized Euclidean distance errors for trajectory prediction accuracy, we also provided Euclidean distance metrics to enable direct comparisons with the existing literature. This supplementary assessment enhances the clarity and comparability of our results. Table 8 demonstrates the superiority of our model in predictive accuracy and forecasting range, establishing it as the most precise UAV trajectory prediction solution to date. In addition to numerical measurements, the model's prediction accuracy was further corroborated using visuals, thus illustrating its capacity to closely match anticipated trajectories with actual pathways in test datasets. This underscores the model's practical utility, affirming its significance for real-world UAV settings.

Table 8. Comparison of our GRU-based UAV trajectory prediction model with existing UAV and aircraft trajectory prediction studies in terms of predictive accuracy, forecasting horizons, and real-time feasibility. The results demonstrate the superior performance of our model across all key metrics, highlighting its effectiveness and adaptability for UAV applications.

Study	Flight Type	Deep Model	Evaluation					
			Evaluation Metric	Real Time?	Comparison Model	MAX Prediction Point (Step)	t + MAX Prediction (RMSE)	t + 1 Prediction (RMSE)
Shu et al. [17] (2021)	UAV	SBULSTM	MSE	-	-	-	-	0.374 m (Mean Distance)
Kant et al. [18] (2022)		LSTM-Autoencoder	MAE, MSE, RMSE	O	Regression, LSTM	-	-	0.1104
Chen et al. [19] (2023)		EC-Bessel-BiLSTM	RMSE	O	Bi-LSTM, LSTM, CNN, BP	1.2 s (6)	Lat: 0.4724 Lon: 0.3509 Alt: 0.4143	Lat: 0.0331 Lon: 0.0271 Alt: 0.0287
Palamas [20] (2023)		LSTM	Euclidean, Precision, Recall, F1-score	O	-	0.3 (3)	(Euclidean) 8.0021	(Euclidean) 2.5521
Ours		GRU	RMSE, MAE, Euclidean	O	Bi-LSTM, LSTM, Bi-GRU, RNN, FlightBERT++	4 s (40)	Lat: 0.000088 Lon: 0.000077 Alt: 2.1593 (Euclidean) 1.5707	Lat: 0.000005 Lon: 0.000002 Alt: 0.1864 (Euclidean) 0.0883
Ma et al. [21] (2020)	Aircraft	CNN-LSTM	RMSE, MAE, MAPE	O	LSTM, BP	5 s (5)	10.192	36.024
Guo D. et al. [22] (2023)		FlightBERT	MAE, RMSE	O	LSTM, Bi-LSTM, Kalman-Filter	300 s (15)	Lat: 0.1067 Lon: 0.1199 Alt: 34.02	Lat: 0.0486 Lon: 0.0558 Alt: 10.59

Table 8. Cont.

Study	Flight Type	Deep Model	Evaluation					
			Evaluation Metric	Real Time?	Comparison Model	MAX Prediction Point (Step)	t + MAX Prediction (RMSE)	t + 1 Prediction (RMSE)
Guo D. et al. [23] (2024)	Aircraft	FlightBERT++	RMSE, MAE, MAPE, MDE	O	LSTM, Transformer, Kalman-Filter, FlightBERT, LSTM + Attention	300 s (15)	Lat: 0.0326 Lon: 0.0265 Alt: 22.89	Lat: 0.0115 Lon: 0.0037 Alt: 12.07
Lan Wu et al. [24] (2024)		FilghtPatchNet	RMSE, MAE	O	LSTM, Bi-LSTM, CNN-LSTM, FlightBERT, FlightBERT++	150 s (15)	Lat: 0.0117 Lon: 0.0157 Alt: 244.34	Lat: 0.00064 Lon: 0.00087 Alt: 123.78

2. Exhibit of Long-Range Predictive Ability

This study illustrated the proposed model's capability to successfully forecast long-term trajectories, in contrast to the numerous prior studies focused solely on short-term forecasts. By progressively augmenting the Forward_Length value, we confirmed the model's ability to predict up to 4 s ahead—an essential function for UAV operations marked by swift directional shifts. At Forward_Length = 40, corresponding to 4 s of prediction, the model achieved RMSE values of latitude: 0.000088, longitude: 0.000077, and altitude: 2.1593, with an average RMSE of 0.7198. These results confirm the model's robustness in handling dynamic and complex flight paths while maintaining high accuracy.

3. Comprehensive Validation of Real-Time Feasibility

Real-time applicability is essential in UAV operations; nevertheless, several existing studies have failed to provide thorough validation of their methodologies within operational restrictions. This work transcended inference latency assessments by examining the model under authentic conditions. Our model was found to execute predictions in 0.0334 s—substantially quicker than the 0.1 s data cycle—while also exhibiting effective resource use. During inference, the model employed 7.56% of CPU resources and 77.78% of GPU resources, maintaining minimal overhead and practical viability for real-time implementation in dynamic UAV contexts. These findings offer compelling evidence for the model's practical utility within operational limits.

4. Exceptional Generalization Performance

A major limitation of prior studies is the absence of evidence for cross-UAV generalization. This study addressed that gap by evaluating the model on two independent UAV datasets—the Synthetic UAV Flight Trajectories and the Package Delivery Quadcopter Drone datasets. Despite being trained on different data, our model maintained high predictive accuracy across both, underscoring robustness and scalability across different platforms and environments. Notably, in testing previously unseen UAV flight logs, it achieved results that were equal to or exceeded the best-performing models—despite those models being tested on data from the same UAV—thereby providing strong evidence of our deep learning model's superior scalability. Its ability to generalize effectively to novel datasets highlights the model's versatility for diverse UAV applications, ranging from controlled research contexts to dynamic operational settings.

This study offers a robust paradigm for UAV trajectory prediction that exceeds the constraints of existing methodologies. The proposed GRU-based model attained the highest accuracy in UAV trajectory prediction studies to date, showcased dependable long-range forecasting abilities, and displayed robust generalization across independent datasets;

furthermore, its real-time feasibility was confirmed through comprehensive resource utilization analyses. These contributions collectively enhance the state-of-the-art in UAV trajectory prediction, positioning our technique as the most precise and pragmatic option for real-world UAV operations.

5. Conclusions

This study introduced a comprehensive framework for UAV trajectory prediction, addressing key challenges and contributing to advancements in the field. The GRU-based model was identified as a strong candidate, achieving robust predictive accuracy and outperforming previous studies on UAV and aircraft trajectory prediction. Our methodology integrates effective generalization capability, reliable long-range forecasting, and practical real-time applicability, positioning it as a scalable solution for real-world UAV applications.

The results indicate that accurate trajectory prediction can contribute to improving the safety, reliability, and scalability of UAV operations. This research provides valuable insights into facilitating collision avoidance, enhancing operational efficiency, and supporting long-range mission planning in complex urban environments. The generalization performance exhibited by the proposed model highlights its adaptability across various datasets, suggesting potential applications in package delivery, emergency response, and anti-drone systems.

Notwithstanding its achievements, this study also delineates other avenues for further research. The model exhibited strong performance across different Forward_Length values; however, visualizations revealed occasional outliers under longer prediction horizons. Subsequent research will concentrate on determining the fundamental reasons for these anomalies and enhancing the model to further reduce such errors. Furthermore, the model's efficacy in altitude forecasting, which showed greater inaccuracies relative to latitude and longitude, necessitates additional examination. This will allow for the creation of sophisticated deep learning systems designed to minimize altitude forecast inaccuracies. This investigation confirmed the model's conceptual feasibility for real-time applications; nevertheless, additional validation in practical circumstances is necessary, and implementing the model on actual UAV systems to evaluate its real-time prediction skills under operational situations will be a vital step. Considering the minimal CPU use noted in this study, subsequent experiments will seek to confirm the model's efficacy in fully integrated UAV environments, thus ensuring its scalability and dependability for forthcoming urban air transportation systems.

Additionally, to further improve the model's generalization performance, future research will incorporate training on a more diverse set of UAV datasets, encompassing various operational conditions, environmental factors, and UAV types. Expanding the dataset scope will enable the model to adapt more effectively to unseen scenarios, thereby enhancing its robustness in real-world applications.

This research presents a robust trajectory prediction system that integrates high predictive accuracy, effective generalization capabilities, and practical real-time applicability. This study addresses key challenges in UAV trajectory prediction, supporting improved safety, efficiency, and scalability in UAV operations, and contributing to the advancement of UAV technology for transportation applications.

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